

Samahith Thellakal

[linkedin](#) | github.com/MowkE

EDUCATION

University of Washington

Electrical and Computer Engineering

Seattle, WA

2026 – 2028

BASIS Independent Fremont Upper

High School

Fremont, CA

2022 – 2026

BIOLOGY

Lead Researcher, Zhang Lab, ASDRP

Biological Chemistry Department

Fremont, CA

Sep. 2022 – Jan. 2025

- **Publication** Coauthored paper in *Bioorganic & Medicinal Chemistry Letters* IF 2.5 “C-19 Silyl and Trityl Ether Analogs of Andrographolide Selectively Inhibit Colon Cancer Cell Growth”
- **Conference Presentations** Presented “Discovery, synthesis, and optimization of 5-phenylisoxazole based covalent inhibitors targeting G12C mutant KRAS for the treatment of cancer” at ACS, SCCUR, and WCBSURC
- **Laboratory Skills** Tissue culture, MTT colony viability assays, scratch migration assays, Western blotting, qPCR
- **Projects**
 - * Combination therapy with Cisplatin and Podophyllotoxin derivatives HCT116, MDA-MB-231
 - * Synthesis and immune analysis of novel Andrographolide analogs
 - * KRAS G12C targeting inhibitors with 5-Phenylisoxazole core structures
 - * Synthesis of C-19 Andrographolide analogs for colon cancer inhibition

Lead Researcher, Pazzi Lab, reThink64

Bioengineering Department

Fremont, CA

Jan. 2025 – Present

- **Laboratory Skills** Lipid Nanoparticle creation, GUVs, MLVs, Fluorescence microscopy, Dynamic Light Scattering DLS, GUV Popping assays
- **MCP Instrumentation Integration** Developed custom Model Context Protocol MCP servers to interface Claude directly with analytical lab equipment including a plate reader, HPLC, LCMS, UV Vis Spectroscopy, and chemistry NMR systems
- **Projects**
 - * Skin mimetic multilamellar vesicle MLV platform for cosmetic delivery
 - * Solvent optimization for improving GUV yield
 - * Carmofur analog membrane rupture analysis using GUV Popping assays

Research Intern, Global Change Research Group (SDSU)

Ecosystem Science Division

San Diego, CA

Jul. 2025 – Sep. 2025

- Evaluated complex ecosystem responses to global environmental changes by vibecoding Python pipelines to process and analyze large scale environmental datasets
- Vibecoded statistical frameworks including n-way ANOVAs, Principal Component Analysis PCA for remote sensing imagery, and Monte Carlo uncertainty modeling to track carbon and water fluxes from soil sensors and eddy covariance towers

Therapeutic Support Specialist Intern, Allé Cauterucci

Support & Education Division

Fremont, CA

Nov. 2022 – Jun. 2023

- Assisted with therapeutic education programs for neurodivergent students developing and implementing custom individualized learning schedules
- Served as a classroom Teaching Assistant TA managing day to day classes and providing operational support

MECHANICAL ENGINEERING

Paid Hardware Intern, Quo Labs @ Founders Inc.

San Francisco, CA

- Assembled wired and validated complex electronic and mechanical systems for **Sam** an emotionally intelligent eldercare robot designed to automate medication cycles and family communication

Octaby (Assistive Robotics Project)

Personal Project

N/A

Apr. 2025 – Present

- Built a camera based assistive wristband using Detectron computer vision and a low latency serial pipeline to translate object locations into haptic feedback for visually impaired users

Hardware Designer & Researcher, Pazzi Lab, reThink64

Fremont, CA

Bioengineering Division

Jan. 2025 – Present

- **Franz Cell Development** Engineered and manufactured a highly efficient custom multiplex Franz diffusion cell setup utilizing vibecad on Onshape and 3D printing to conduct high throughput transdermal drug delivery testing
- **Cost & Throughput Optimization** Reduced transdermal testing hardware costs from \$500 to \$1.56 while scaling testing throughput from 1 to 96 parallel trials manuscript in prep for *Nature BME* IF 26.3
- **Materials Recycling** Coordinated with 3D printing manufacturers to recycle thousands of waste polypropylene pipette boxes from the laboratory into active 3D printer filament for structural components

CONFERENCES

- Y Combinator Startup School 2026, ACS Conference, SCCUR, WCBSURC

SOFTWARE ENGINEERING & MARKETING

Outreach Lead / Programmer, FRC Team 9470

Fremont, CA

Software & Operations Division

Mar. 2024 – Present

- Programmed real time shoot on the move kinematic calculations and autonomous navigation routines in **Java** using the **WPILib** framework
- Helped map a multi variable shooter lookup table shooter map to dynamically scale motor velocity and launch angles based on live distance tracking data
- Secured \$7,000 in funding from grants and selling merch, coordinated match play across alliances, and cofounded neurodiversity initiatives to present quiet rooms and sensory tools at the World Championships
- Coordinated alliance match strategy to secure a division finalist finish ranking #2 at our division at the FRC World Championships advancing to the Einstein Field as one of the top 16 teams

Monkeyless (Mindfulness Screen Time App)

Remote

Lead Developer / Marketer

Jan. 2026 – Present

- Engineered and launched features on iOS and Android to stop screen addiction enforcing a custom mindfulness breathing block using native system level hooks and background process permissions
- Shipped a production ready MVP to the App Store in under 4 weeks of rapid AI augmented development in React Native
- Collaborated with the startup team to iterate on new UI features based on user feedback to manage ADHD symptoms
- Led social media marketing campaigns scaling adoption to 5,000+ organic downloads and generating 144k+ views on Instagram

OpenTrade (YCS 2026)

San Francisco, CA

Growth Outreach & Front-End Design

2026

- School outreach campaign sending 1,000+ emails to educators and institutions for product demonstrations in India and Singapore
- Contributed to front-end product design by developing UI layouts, visual components, and overall product usability

BioSim (Passion Project)

N/A

Founder / Lead Developer

Oct. 2022 – Present

- Architected and deployed a web based 3D biological simulation platform using **Babylon.js** and WebGL rendering cellular organ systems and atomic systems directly in the browser
- Designed and optimized an interactive 3D Periodic Table displaying dynamically rendered spatial structures for all 118 elements expanding interactive modeling tools to include AP Chemistry curriculums
- Accelerated asset creation cycles by 10x using custom prompt pipelines to generate WebGL coordinate math
- Scaled platform reach from 6 local schools to 79 school districts across 98 countries through targeted educator email outreach and SEO optimization
- Project Links bio-sim.us and bio-sim.us/chemistry/

Software Engineering Contractor, ShogunAI

Remote

AI Product Engineering

2026, one-week contract

- Independently rebuilt the end-to-end living knowledge-base workflow of GodHands and integrated it into ShogunAI's local-first desktop agent, preserving evidence-linked context across people, projects, decisions, and handoffs
- Delivered a working integration in one week, adapting knowledge ingestion, retrieval, and user-interface flows to ShogunAI's existing product architecture